NETWORK AND PHYSICAL SECURITY ASSIGNMENT

1. Create a table listing 5 common network threats and give one real-world example from an app or service such as WhatsApp, Instagram, or Paytm.

Network threat	Example involving an app or service	Possible impact
1. Phishing	A fake Instagram login message tricks a user into entering their password on a look-alike website.	An attacker may take over the account.
2. DDoS attack	Attackers send huge amounts of traffic towards an online food-ordering service such as Zomato.	The service may become slow or unavailable.
3. Man-in-the-middle (MITM)	On unsecured public Wi-Fi, an attacker attempts to intercept traffic between a user and an app.	Sensitive information may be exposed if connections are not properly encrypted.
4. Malware	A user downloads a fake WhatsApp update containing malicious software.	The malware may steal data or damage the device.
5. Password attack	An attacker tries reused or commonly guessed passwords to access a Paytm account.	Unauthorized access may lead to account misuse or financial loss.

2. Simulate a brute-force attack scenario by writing a Python script that tries every possible 4-digit PIN against a preset value. Print the number of attempts required.

```
secret_pin = "4826"
attempts = 0

for number in range(10000):
    guess = f"{number:04d}"
    attempts += 1

    if guess == secret_pin:
        print("PIN guessed:", guess)
        print("Number of attempts:", attempts)
        break
```

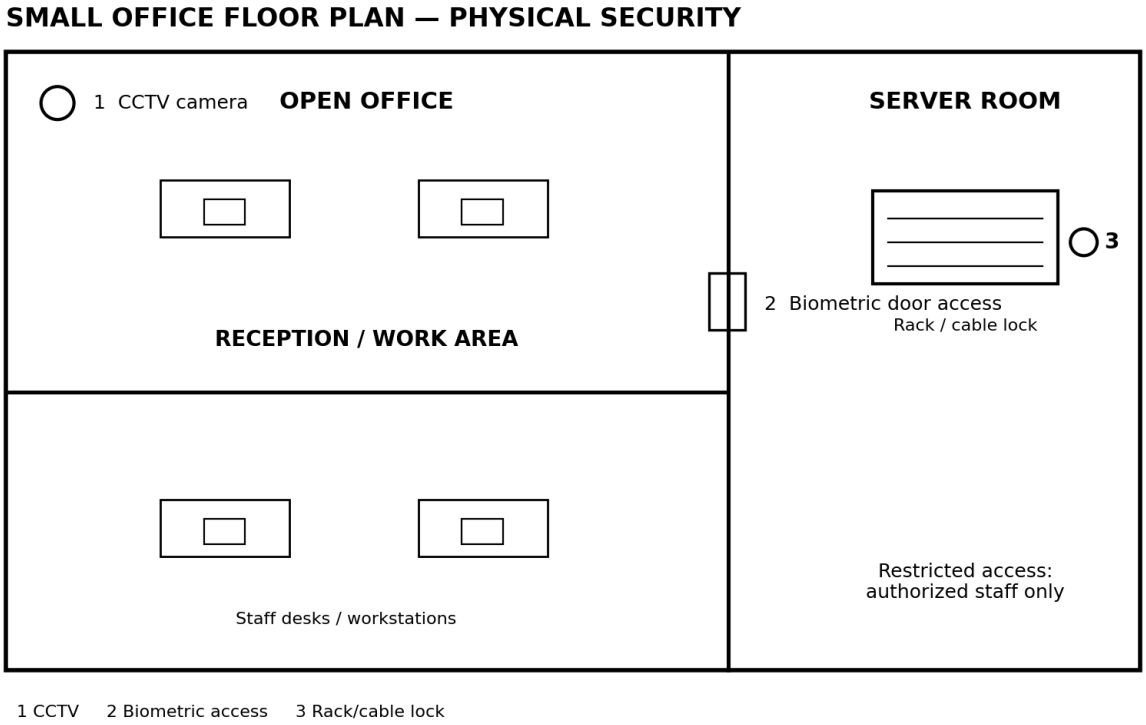
Explanation: The loop checks every PIN from 0000 to 9999 in order. The attempts counter increases each time a PIN is tested. When the guessed PIN matches the preset PIN, the program prints the correct PIN and total attempts, then stops.

Expected output: PIN guessed: 4826
Number of attempts: 4827

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each.

Physical security risk	Possible consequence	Mitigation step
Unauthorized entry	An intruder could steal, damage, or tamper with servers and network devices.	Use biometric access or access cards and maintain entry logs.
Fire or smoke	Equipment and stored data may be damaged, causing service downtime.	Install smoke detectors and suitable fire-suppression systems.
Power failure	Servers and network equipment may shut down unexpectedly.	Use UPS units and backup power generators.
Water leakage or flooding	Water may damage electrical equipment and cause outages.	Keep equipment away from water pipes and install leak detectors.

4. Draw a simple floor plan of a small office and mark at least 3 physical security measures to protect network equipment.



Security measures shown:

- 1. CCTV camera: monitors the office entrance and helps record suspicious activity.
- 2. Biometric door access: restricts entry to the server room to authorized staff.
- 3. Rack/cable lock: helps prevent unauthorized removal of network equipment.

5. Generate a checklist of 7 actions a network administrator should take to mitigate network threats and physical security breaches, adapted to a college computer lab.

- Keep lab computers, routers, switches, and firewall software updated to fix known security weaknesses.
- Use strong, unique passwords for administrator accounts and enable multi-factor authentication wherever available.
- Configure the lab firewall to block unnecessary inbound and outbound ports, and review its rules regularly.
- Install and update endpoint protection on lab computers; scan removable USB drives before use.
- Restrict the server/network cabinet and equipment room using keys or access cards, and allow entry only to authorized staff.
- Install CCTV near lab entrances and network equipment, and review recordings when an incident is reported.
- Back up important lab configurations and college data regularly, store backups securely, and test that they can be restored.

Review and adaptation: This checklist is tailored to a college computer lab, where many students use shared computers and removable drives. It combines technical controls (updates, passwords, firewall rules, endpoint protection, and backups) with physical controls (restricted equipment access and CCTV). Lab staff should report suspicious activity promptly and follow the college's incident-reporting procedure.

Conclusion: Network security and physical security work together. Protecting devices, accounts, rooms, and network traffic reduces the chance of unauthorized access, data loss, and service disruption.